

ECE 647: Lecture 1

Introduction

IN ECE547, the goal is often to analyze a system (e.g. a queueing system), or to analyze a protocol or an algorithm (e.g. a priority queueing discipline, Bellman-Ford algorithm, or CSMA/CD)

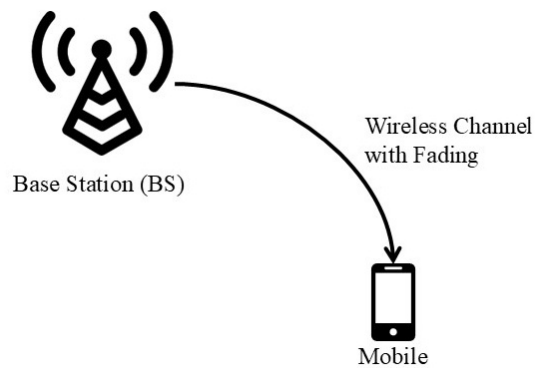
While performance analysis is of course crucial, for many research problems you may not have been given an algorithm or a protocol to begin with.

In other words, you are free to design your own algorithm/protocol.

You then wonder how to **design an algorithm/protocol** that is in certain sense the "**best**".

Examples

Transmission on a wireless channel with fading:

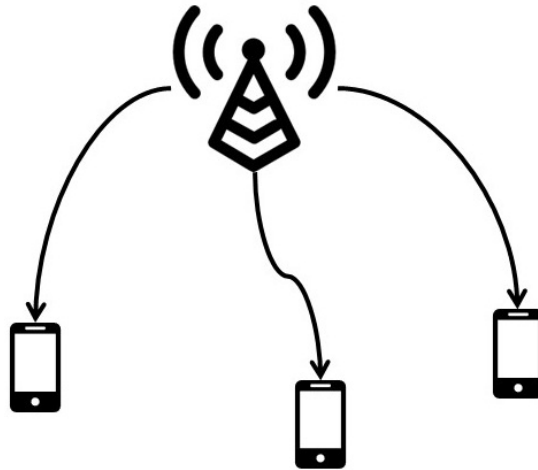


Due to mobility and fading, channel gains between the base station and the mobile changes with time.

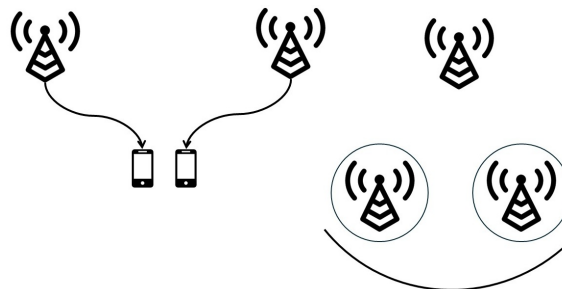
What should the transmission power at the BS be when the channel gain is g ?

- Fixed power?
- Variable power to maintain fixed rate?
- What is the "best" power assignment?

What if the BS is transmitting to multiple users at the same time?



What if there are multiple BS or femto-cell?



Optimization

In the first part of the class we will study **convex optimizations**, which will be a useful tool to answer such questions.

This process normally takes the following **high-level steps**:

1. Built a **system model**

- $g(t)$: channel gain, randomly distributed with some distribution.
- $r(t) = \log \left(1 + \frac{g(t) \cdot p(t)}{N} \right)$

2. Identify the **objectives** to optimize

- maximize transmission rate, or
- minimize transmission power

3. Identify the **constraints**

- an upper bound on the transmission power
- a lower bound on the transmission rate

without proper constraints, the optimization problem could become trivial.

4. Formulate the **optimization problem**.

Assume $g(t)$ is i.i.d. (independent and identically distributed), from a finite set of values $\{g_1, g_2, \dots, g_M\}$.

Let $q_k = Pr \{g(0) = g_k\}$

Let p_k denote the power assignment when the channel gain is g_k .

Then the optimization problem is formulated as:

$$\begin{aligned} & \max_{p_1, p_2, \dots, p_M} \sum_{k=1}^M q_k \log \left(1 + \frac{q_k p_k}{N} \right) \\ & \text{subject to } \sum_{k=1}^M q_k p_k \leq p_{\max} \end{aligned}$$

Q: Is this the only possible formulation?

A: No. Can instead minimize power, delay, etc. However, there is a reason why this is one of the more popular ways to formulate the problem, as we will point out soon.

5. Find the **solution** to this problem

Such a solution may provide a direct relationship between $\{p_1, \dots, p_M\}$ and the modeling parameters.

$$\vec{p}^* = f(\vec{g}, \vec{q})$$

At other times, a direct relationship may not be possible or convenient

- Not all equations have closed form solutions

In this case, we may be happy with an iterative solution

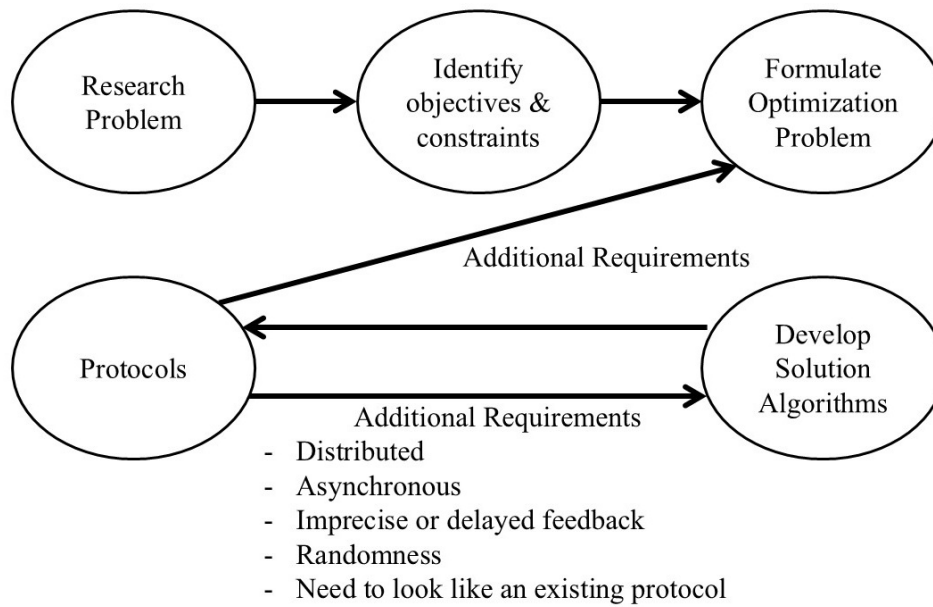
$$\begin{aligned}\vec{p}(t+1) &= f(\vec{p}(t) \mid \vec{g}, \vec{q}) \\ \vec{p}(t+1) &\rightarrow \vec{p}^*\end{aligned}$$

Be it direct solutions or iterative solutions, for practical reasons we often

- **prefer online adaptive solutions** over offline solutions
 - estimating model parameters can be time-consuming
 - or inaccurate
 - or model parameters can change over time
- **prefer distributed solutions** over centralized solution
 - single point of failure
 - lack of scalability

An online distributed solution can potentially be turned into a practical protocol that can be used in real systems.

In Summary



Convex Problems